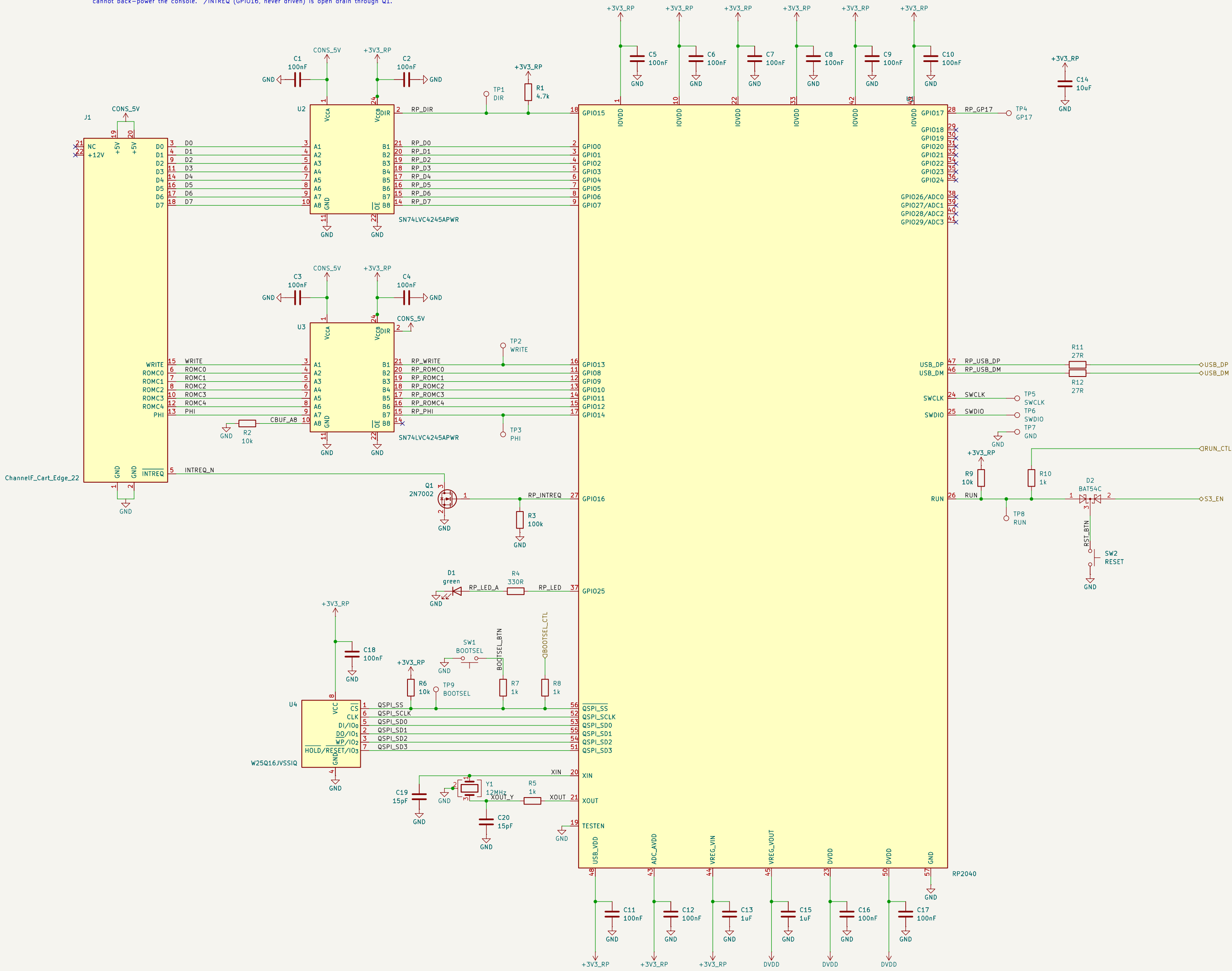
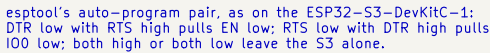


The Fairchild Channel F Videocart edge, 22 contacts (channelf.as-veswiki Pinouts): the F8 bus has NO address bus -- D0-D7, five ROMC lines, WRITE (one machine cycle per pulse) and PHI -- and no reset line (a console reset arrives as ROMC 0B). +12V (22) and NC (21) are left open.
U2 / U3 SN74LVC4245A: A port on CONS_SV (TTL inputs, VIH 2.0 V; drives 5 V levels -- the F3950 data bus wants VIH 2.9 V, Table 7). B port on +3V3_RP (the RP2040 is not 5 V tolerant). DIR high = A -> B = console -> cart.
U2, data: DIR = GPIO15 (channelf_cart.h: 1 = console -> cart, 0 = cart drives D0-D7). R1 holds it high through reset and boot against the pad pull-down. U3, control: DIR tied high. CONS_SV comes straight off the edge, ahead of the OR diode; with the console off both translators go high-Z (VCC isolation), so a USB-powered cart cannot back-power the console. /INTREQ (GPIO16, never driven) is open drain through Q1.



Supplies (Hardware design with RP2040, 2.1): IOVDD x6, USB_VDD, ADC_AVDD and VREG_VIN on +3V3_RP, the AP2112K LDO: the on-chip regulator makes DVDD [1.1 V] at VREG_VOUT for both DVDD pins. 100 nF at every supply pin, 1 uF at VREG_VIN and at VREG_VOUT (2.1.3). Crystal: ABM8-272-T3, 15 pF each side, 1k on XOUT (2.3). Flash: W25Q16JV, 2 MB (fujichannelf.cmake) /CS pulled up (2.2). USB: 27R at the chip (2.4); the S3 is the host. RESET resets the RP2040 and the ESP32-S3 together (Q2 steers RST_BTN onto RUN and S3_EN) -- the way back to CONFIG after a booted Videocart. Hold BOOTSEL while pressing RESET for the RP's USB boot ROM: the S3 does both itself; IO4 (RUN_CTL) low resets the RP; IO5 (BOOTSEL_CTL) low through that reset selects BOOTSEL. First flash: SWD pads (the RP has no PC-facing USB).



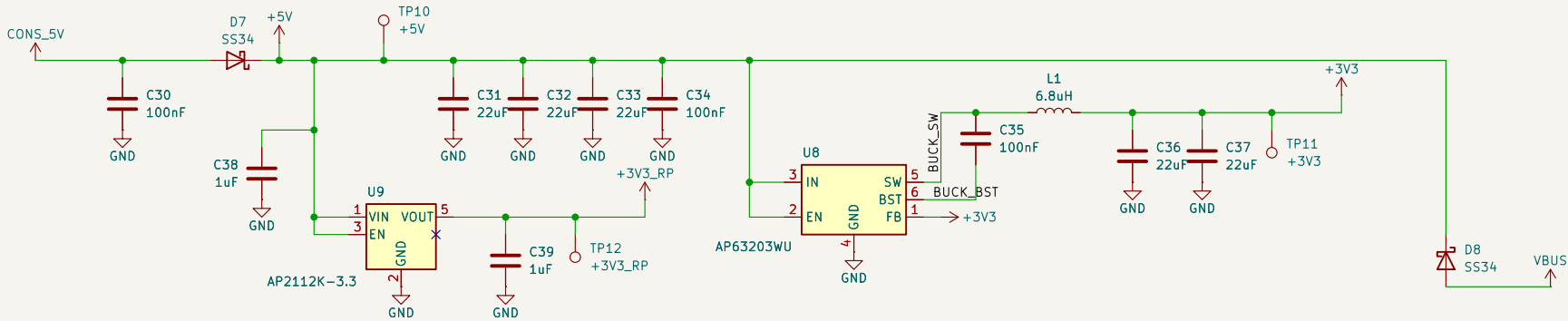
Firmware: fujinet-firmware pico/channelf (as-is) + fujiuniversal-channelf
CERN-OHL-W-2.0 (S3/USB/power from FujiNet-Astrocade-Rev0 and its sources)
Generated from tools/design.py + tools/sch_layout.py – edit those, not this file
FujiNet

Sheet: /usb/
File: usb.kicad_sch

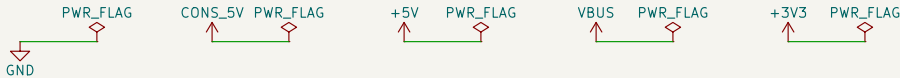
Title: FujiNet Channel F Rev0 – USB-C + CP2102N

Size: A3	Date: 2026-10-09	Rev: 0
KiCad E.D.A. 10.0.7		Id: 4/5

Power tree: +5V is the OR of two sources, so neither can back-feed the other.
CONS_5V edge pins 19 / 20, the console's +5V, through D7 SS34.
VBUS USB-C, through D8 SS34.
SS34 VF ~0.35 V at the cart's ~0.4 A; +5V is 4.4–4.9 V. The only 5 V logic is the two SN74LVC4245A A ports (cart-rp sheet), and they take CONS_5V itself, ahead of the diode: off with the console, so the translators isolate the bus whenever the console is off.
U9 AP2112K-3.3 LDO -> +3V3_RP: the RP2040 (IOVDD, USB_VDD, ADC_AVDD, VREG_VIN), its flash and both translators' B ports; in dropout it follows +5V up as the console rail rises.
U8 AP63203 3.3 V / 2 A buck -> +3V3: ESP32-S3 (WiFi bursts ~400 mA), CP2102N, microSD; it waits for UVLO, then soft-starts over 4 ms. TP10, TP11 and TP12 are bring-up test points. Console +5V headroom under WiFi bursts is unmeasured: power the cart from USB-C if the console rail sags (bring-up checklist, README).



PWR_FLAG: these rails reach their loads only through passive parts (the edge pins, USB-C, the SS34s, the buck's inductor).



Firmware: fujinet-firmware_pico/channelf (as-is) + fujiversal-channelf
CERN-OHL-W-2.0 (S3/USB/power from FujiNet-Astrocade-Rev0 and its sources)
Generated from tools/design.py + tools/sch_layout.py – edit those, not this file
FujiNet

Sheet: /power/
File: power.kicad_sch

Title: FujiNet Channel F Rev0 – Power

Size: A3	Date: 2026-10-09	Rev: 0
KiCad E.D.A. 10.0.7		Id: 5/5